

DOHA INTL, Qatar

WMO: 411700

Lat: 25.261N

Lon: 51.565E

Elev: 35

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.2	56.8	30.2	24.4	82.9	34.2	28.9	82.7	21.0	67.5	19.4	67.8	7.5	300	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	111.7	72.2	109.6	72.5	107.6	73.4	88.1	95.6	87.3	95.2	86.4	94.9	13.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
86.3	192.5	93.4	85.6	188.5	93.4	84.3	180.2	93.3	53.4	96.2	52.5	94.6	51.3	95.2	93.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	20.1	17.9	DB	50.9	117.0	2.7	1.7	48.9	118.2	47.3	119.2	45.8	120.2	43.8	121.4
			WB	43.9	89.8	3.1	1.0	41.7	90.6	39.8	91.2	38.1	91.8	35.8	92.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.8	65.7	67.7	73.6	82.4	92.6	97.2	98.8	97.5	93.9	87.8	78.2	69.5	(6)
	DBStd	12.62	4.20	4.44	4.72	4.94	4.40	3.86	3.26	3.04	3.20	3.35	4.51	4.40	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	73	41	19	1	0	0	0	0	0	0	0	0	12	(9)
	CDD50	12345	486	496	731	971	1321	1416	1512	1474	1316	1171	847	604	(10)
	CDD65	6943	62	95	267	521	856	966	1047	1009	866	706	397	151	(11)
	CDH74	98639	116	279	1559	5659	13066	16063	17721	16780	13789	9751	3372	484	(12)
	CDH80	65099	10	50	445	2588	8682	11747	13257	12316	9474	5431	1049	50	(13)
(14) Wind	WSAvg	8.4	8.0	9.2	9.5	9.1	9.4	10.0	8.9	7.7	7.4	6.8	7.7	7.5	(14)
(15) Precipitation	PrecAvg	2.8	0.5	0.7	0.8	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.5	(15)
	PrecMax	10.3	4.0	5.1	5.6	1.6	0.4	0.0	0.0	0.0	0.0	0.7	4.3	6.1	(16)
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.4	0.8	1.0	1.3	0.3	0.0	0.0	0.0	0.0	0.0	0.1	0.7	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.7	85.9	94.5	102.5	111.4	115.0	114.9	113.3	109.2	103.9	93.6	84.4	(19)
		MCWB	63.1	63.6	65.6	68.0	69.8	71.2	72.1	74.0	73.3	71.5	68.5	65.9	(20)
	2%	DB	76.9	80.3	88.2	97.2	107.7	111.4	111.6	109.7	105.9	100.2	90.0	80.9	(21)
		MCWB	63.4	62.8	64.6	67.3	70.5	71.5	71.9	75.4	74.6	72.9	71.0	67.5	(22)
	5%	DB	74.8	77.1	84.4	94.7	104.4	109.2	109.5	107.4	103.6	97.1	87.8	78.8	(23)
		MCWB	63.4	62.7	63.3	67.1	70.2	71.2	72.6	77.1	75.9	74.6	71.6	66.7	(24)
	10%	DB	72.3	74.8	81.2	91.4	102.0	106.2	107.3	104.4	100.8	94.8	85.8	76.7	(25)
		MCWB	62.7	62.5	63.3	67.3	70.0	71.6	73.9	79.4	78.3	75.7	71.3	66.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.8	69.5	72.0	76.0	81.9	85.6	89.0	89.7	87.7	84.1	78.6	73.6	(27)
		MCDB	73.2	74.6	78.8	85.5	89.4	92.7	96.1	96.4	95.9	92.2	84.5	77.6	(28)
	2%	WB	68.2	68.1	70.5	74.4	80.1	83.9	87.6	88.5	86.4	82.3	76.6	71.5	(29)
		MCDB	72.3	73.3	77.4	84.6	90.0	92.6	95.0	95.8	94.8	90.0	84.0	76.6	(30)
	5%	WB	66.5	67.0	69.3	73.0	78.3	82.6	86.7	87.8	85.4	81.0	75.2	69.8	(31)
		MCDB	71.4	72.4	76.4	83.9	90.5	92.2	94.5	95.4	94.0	89.3	83.1	75.3	(32)
	10%	WB	64.9	65.6	68.1	71.9	76.6	81.0	85.5	87.1	84.5	79.9	73.8	68.4	(33)
		MCDB	70.6	71.5	75.8	83.3	90.5	92.4	94.2	95.3	93.5	88.7	81.8	74.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.7	13.6	15.1	16.8	18.7	19.1	17.1	15.6	15.6	15.0	12.7	12.7	(35)
		MCDBR	14.5	16.2	19.5	21.0	21.6	21.8	20.5	19.4	18.6	18.0	14.7	14.2	(36)
	5% WB	MCWBR	6.9	7.1	8.2	9.0	10.3	11.1	11.8	11.4	10.7	9.9	7.4	6.7	(37)
		MCWBR	12.1	12.5	14.7	17.6	19.8	18.8	15.8	14.6	14.5	15.4	13.6	12.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.503	0.551	0.617	0.664	0.654	0.630	0.686	0.647	0.569	0.511	0.505	0.483	(40)
		taud	1.892	1.782	1.644	1.593	1.595	1.636	1.592	1.673	1.800	1.916	1.936	1.955	(41)
	Ebn at Noon		232	231	224	218	220	224	211	219	233	239	229	231	(42)
		Edh at Noon	56	66	80	86	86	82	85	78	68	57	53	50	(43)
(44) All-Sky Solar Radiation	RadAvg		1185	1430	1650	1840	2098	2206	2091	2005	1857	1588	1246	1106	(44)
	RadStd		71	76	102	130	57	58	54	46	36	39	59	72	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.17	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	+457	+430	(46)
(47)	Regional (3 neighbors)	+1.08	N/A	-2.66	+1.35	N/A	N/A	N/A	N/A	+428	+403	(47)

Nomenclature: See separate page